

Measuring Chain of Thought Faithfulness

Based on *Measuring Chain of Thought Faithfulness by Unlearning Reasoning Steps*^{*}
Group 27 — Course Project Report

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Abstract

Chain-of-Thought (CoT) prompting has substantially improved the reasoning performance of large language models, yet whether generated rationales faithfully reflect the model’s internal parametric reasoning remains an open question. This paper reproduces and extends *Measuring Chain of Thought Faithfulness by Unlearning Reasoning Steps*, focusing on the Parametric Faithfulness Framework (PFF) and its instantiation through Faithfulness by Unlearning Reasoning steps (FUR). We audit the English reproduction against the available source-result files. Across eight model–dataset pairs for which the two runs use the same learning rate, question-level FF-HARD has a mean absolute difference of 2.1 percentage points (maximum 4.8); across seven paired comparisons with different learning rates, the mean absolute gap rises to 35.5 points. Case-level analysis further shows that close aggregate rates need not arise from the same flipping steps. We then examine Chinese reasoning with Qwen models on C-Eval and explore broader LoRA-based intervention, selective unlearning, Tree-of-Thought reasoning, and additional diagnostic objectives. The results support FUR as a useful causal probe, while showing that its measured faithfulness signal depends materially on optimization strength, intervention scope, and dataset characteristics.¹

1 Introduction

Chain-of-thought (CoT) prompting elicits intermediate rationales from large language models and often improves performance on arithmetic, commonsense, and scientific reasoning tasks (Wei et al.,

2022; Kojima et al., 2022; Nye et al., 2021). Related decoding methods such as self-consistency further show that sampling multiple reasoning traces can improve final answers even when any single trace is noisy (Wang et al., 2023). These empirical gains have made CoT a common interface for inspecting model reasoning, but they also create a sharper interpretability question: *are the visible rationales causal descriptions of how the model reaches its answer, or fluent explanations generated alongside an answer that was already supported elsewhere in the model?*

This question is usually framed as **faithfulness**. A faithful rationale should track the information and computation actually used by the model, not merely provide a plausible explanation that a human reader accepts (Jacovi and Goldberg, 2023). Several studies have shown that CoT explanations can be sensitive to prompt framing, post-hoc rationalization, or misleading context (Lanham et al., 2023; Turpin et al., 2023; Lyu et al., 2023). Input-space perturbations, for example deleting or corrupting a rationale sentence and observing whether the answer changes, are useful but incomplete: they intervene on text in the prompt, not on the parametric knowledge that generated the rationale. A model may change its answer because it sees a different context, or keep the same answer because the answer is recoverable from other cues, without revealing whether the original rationale was part of the model’s internal decision process.

Tutek et al. (2025) address this gap with the **Parametric Faithfulness Framework (PFF)**. Rather than perturbing the prompt, PFF asks whether removing from the model parameters the information expressed in one reasoning step changes the direct answer. Its concrete method, **Faithfulness by Unlearning Reasoning steps (FUR)**, uses machine unlearning through Negative Preference Optimization with KL regularization (NPO+KL) (Zhang et al., 2024) to suppress one

^{*}You can access the reproducible version of the paper at the following URL: <https://arxiv.org/abs/2502.14829>

¹The project repository is available at <https://github.com/KouseiAimer/Measuring-Faithfulness-Group27>. A mirror is stored at [cfff:/inspire/hdd/project/fdu-aidake-cfff/public/llm-26-final-projects/group-27](https://cfff.inspire.hdd/project/fdu-aidake-cfff/public/llm-26-final-projects/group-27).

CoT step while preserving behavior on held-out examples. If the answer flips after the step is unlearned, the step supplies evidence of parametric faithfulness. This turns faithfulness into a controlled causal test with three simultaneous requirements: the target step must be erased, unrelated behavior must remain stable, and the answer distribution must be re-evaluated without re-providing the CoT.

We reproduce and extend PFF/FUR along the experimental path used in our project presentation: first reproducing the original English MCQA setting, then moving to Chinese C-Eval reasoning, and finally exploring scalable, structural, and training-based extensions. The main contributions are:

Faithful reproduction: We evaluate four instruction-tuned models over five MCQA benchmarks and explicitly audit result comparability. The audit separates eight exact-paper-LR pairs from seven pairs whose result files use different learning rates. The exact pairs reproduce closely at the aggregate level (2.1-point mean absolute FF-HARD gap and 1.0-point FF-SOFT gap), whereas the LR-mismatched pairs differ by 35.5 FF-HARD points on average. Case-level analysis shows that aggregate agreement does not imply identical flipping steps.

Chinese-language extension: We extend FUR to Chinese reasoning with Qwen3-8B and Qwen3-4B on C-Eval (Huang et al., 2023). Although unlearning is effective and specific, only 9.13% of Qwen3-8B instances exhibit hard answer flips, indicating a sparse causal connection between visible Chinese CoT steps and final predictions in the evaluated Qwen3 + C-Eval setting.

LoRA-based lightweight intervention: We replace full-parameter updates with LoRA adapters and show that expanding adapter targets from MLP down-projections to all linear layers increases question-level FF-HARD from 12.6% to 44.3% while maintaining 99.1% specificity.

Selective FUR for cost reduction: We show that unlearning a single selected step (top-1) reduces computational cost by over 80% relative to full-stepwise FUR while recovering approximately 40% of the faithful instances identified by the full method.

Tree-of-Thought faithfulness: We generalize PFF from linear CoT to Tree-of-Thought (ToT) reasoning and find that multi-path deliberation reveals stronger parameter dependence on ARC-Challenge (+3.4 FF-HARD, +7.3 FF-SOFT).

Additional diagnostic objectives and distilled

CoT: We conduct first- k forgetting, representation-loss, and Causal-CoT diagnostics, and we test one supervised fine-tuning pass on DeepSeek-revised CoTs. These extensions improve our understanding of what FUR measures: compact or mechanistically altered rationales do not automatically yield stronger hard faithfulness.

2 Background: Parametric Faithfulness Framework

2.1 CoT Faithfulness

Given a language model π_θ and a query x , let $y = \pi_\theta(x)$ denote the direct, no-CoT answer prediction, and let $r = (r_1, r_2, \dots, r_k)$ denote a generated rationale segmented into k reasoning steps. Classical explanation faithfulness asks whether r reflects the computation that produces y (Jacovi and Goldberg, 2023). For CoT, the problem is sharper because the explanation is produced by the same model that answers the question. CoT can improve accuracy (Wei et al., 2022; Kojima et al., 2022), yet generated rationales may still be rationalizations or prompt-compliance artifacts (Turpin et al., 2023; Lanham et al., 2023).

The PFF view narrows the question to parametric dependence: *does the parametric knowledge expressed in step r_i causally support the direct answer y ?* This does not require every generated sentence to be faithful, nor does it claim to recover a full internal circuit. Instead, it asks whether a targeted parameter intervention on one stated reasoning step changes the answer under controlled conditions.

2.2 Two-Stage Framework

PFF (Tutek et al., 2025) operationalizes faithfulness measurement through two stages:

Stage 1: Parameter Intervention. For each reasoning step r_i , apply a parameter-level intervention Φ_i that removes the information contained in r_i from the model parameters θ , yielding intervened parameters $\theta^{(i)}$. The intervention must satisfy three validity criteria:

- **Efficacy:** The probability of generating r_i under $\pi_{\theta^{(i)}}$ should be substantially reduced relative to π_θ .
- **Specificity:** Behavior on unrelated examples should be preserved, i.e., predictions on a held-out set should remain unchanged.

- **General Capability:** Broad task capabilities (e.g., as measured by MMLU) should not degrade significantly.

Stage 2: Faithfulness Evaluation. Compare the direct answer prediction before and after intervention. If $\pi_{\theta(i)}(x) \neq \pi_{\theta}(x)$, then step r_i is considered *parametrically faithful*: the answer causally depends on the parametric knowledge that produced r_i .

Two complementary metrics quantify this:

- **FF-HARD:** A binary indicator of whether *any* unlearned step changes the predicted answer class (argmax flip).
- **FF-SOFT:** A continuous measure of the mean reduction in probability mass assigned to the original answer after unlearning.

The paper-protocol FF-HARD is usually computed on examples where direct and CoT-conditioned answers agree before intervention. This filter avoids attributing an answer to a rationale that did not explain the model’s initial direct prediction.

2.3 FUR: Instantiation via Machine Unlearning

FUR instantiates PFF using machine unlearning as the parameter intervention. The core technical components are:

Forget and Retain Sets. For a target reasoning step r_i , the forget set $\mathcal{D}_{\text{FG}}$ consists of the content words (nouns, verbs, adjectives, numerals) from r_i . The retain set $\mathcal{D}_{\text{RT}}$ consists of content words from unrelated CoT steps so that unlearning remains localized to the target step.

NPO + KL Objective. FUR uses Negative Preference Optimization (NPO) (Zhang et al., 2024) with KL regularization to erase the target step:

$$\mathcal{L}_{\text{NPO}} = -\frac{2}{\beta} \mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{FG}}} \left[\log \sigma \left(-\beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} \right) \right] \quad (1)$$

$$\mathcal{L} = \mathcal{L}_{\text{NPO}} + \lambda \mathcal{L}_{\text{KL}} \quad (2)$$

where π_{ref} is the frozen reference model (the pre-unlearning checkpoint), β controls the strength of preference optimization, and $\mathcal{L}_{\text{KL}}$ is a KL divergence term that limits divergence on the retain set.

Stepwise Unlearning Protocol. For each question instance, FUR:

1. Generates a CoT through two-step prompting (generate reasoning chain $\rightarrow$ answer based on chain).

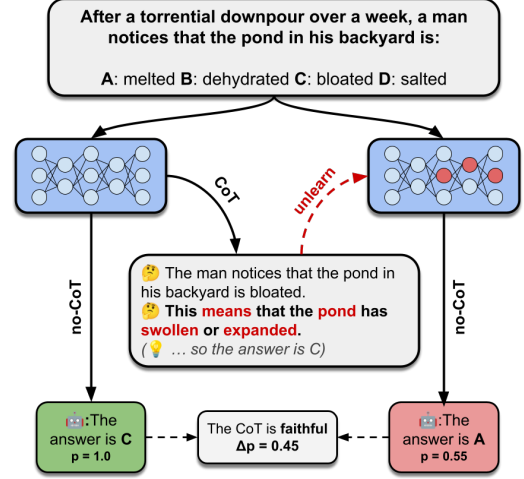


Figure 1: FUR measures faithfulness by unlearning one reasoning step at a time and testing whether the direct answer changes. The intervention must suppress the target step while preserving unrelated behavior.

2. Segments the CoT into sentence-level steps using NLTK.
3. For each step i , performs a separate unlearning run that targets only the content words in that step.
4. Evaluates the direct (no-CoT) answer probability distribution before and after unlearning.
5. Records whether the argmax prediction flips (FF-HARD) and by how much the original answer probability drops (FF-SOFT).

Why FFN down-projection matters. The original FUR implementation updates only the MLP/FFN down-projection subspace, motivated by the hypothesis that factual and reasoning knowledge is partly localized in feed-forward layers. Our extensions test this assumption by using LoRA (Hu et al., 2022) to broaden the intervention scope and by comparing behavior under all-linear adapters, selective step choice, and multi-path ToT rationales (Yao et al., 2023).

3 Reproduction Experiments

3.1 Experimental Setup

Models. We evaluate four instruction-tuned models from three model families, all under a zero-shot setting: LLaMA-3-8B-Instruct, LLaMA-3.2-3B-Instruct (Meta LLaMA family), Mistral-7B-Instruct-v0.2 (Mistral AI), and Phi-3-mini-4k-Instruct (Microsoft).

Comparison category	Groups
Exact paper LR, paired	8
Paper LR, source unavailable	4
Different LR, paired	7
Incomplete/non-comparable	1

Table 1: Audit of the 20 intended English reproduction groups. The incomplete group is Sports/LLaMA-3.2-3B, whose local mainline output contains only one question.

Datasets. We use five multiple-choice QA benchmarks: ARC-Challenge (Clark et al., 2018) (scientific reasoning), OpenBookQA (Mihaylov et al., 2018) (elementary science commonsense), StrategyQA (Geva et al., 2021) (implicit multi-hop logic), TruthfulQA (Lin et al., 2022) (factuality and misconceptions), and BigBench-Hard Sports from BIG-bench (Srivastava et al., 2023) (sports-domain understanding). Each dataset contributes up to 250 CoTs (248 for Sports), of which 20 are held out for specificity. Complete mainline runs therefore contain 228–230 target questions; the reproduction audit requires at least 220 unique target IDs.

Unlearning Configuration. We apply sentence-level stepwise NPO+KL for five epochs, target content words (nouns, verbs, adjectives, and numerals), and restrict parameter updates to the FFN/MLP down-projection layer. Learning rates are dataset-model specific rather than model-wide constants; the registry values are listed in Appendix 12. All direct reproduction claims below are restricted to pairs for which both our run and the available source run use the same registry learning rate.

Comparison Audit. The 20 intended dataset-model groups separate into four categories: eight exact-paper-LR paired groups, seven paired groups whose result files use different learning rates, four complete but unpaired runs, and one incomplete local mainline group (Sports/LLaMA-3.2-3B). We keep these categories separate rather than imputing missing values or mixing non-comparable configurations.

3.2 Paper-Level Reproduction

Table 1 summarizes the comparison coverage. Figure 2 reports question-level FF-HARD for the eight directly comparable pairs and includes the four remaining paper-LR runs for completeness.

Aggregate agreement. Across the eight exact-paper-LR source pairs, our FF-HARD values closely track the source results (Pearson $r =$

0.976). The mean absolute gap is 2.07 percentage points, the largest absolute gap is 4.82 points, and the signed mean gap is 0.33 points. FF-SOFT is similarly stable, with a mean absolute difference of 0.96 points. The directly paired local FF-HARD values range from 28.9% to 71.7%, showing substantial dataset and model variation even when the reproduction itself is close.

Scope of the claim. The evidence supports aggregate reproducibility for the eight controlled pairs. It does not establish instance-level replication: similar FF-HARD totals can be produced by different questions or reasoning steps crossing the decision boundary. Appendix Figure 14 examines this distinction directly.

3.3 Learning Rate Sensitivity

Here, an “LR-mismatched” comparison means that our result file and the available source-result file use different learning rates. This classification is file-pair specific: in one case (OpenBookQA/LLaMA-3.2-3B), our run uses the registry paper LR while the available source file uses a lower LR.

Across these seven pairs, the mean absolute FF-HARD gap is 35.55 percentage points, compared with 2.07 points under exact learning-rate matching. The signed gaps range from -40.43 points (OpenBookQA/Phi-3) to $+61.74$ points (StrategyQA/LLaMA-3-8B).

The low-LR Phi-3 runs illustrate under-unlearning. On ARC-Challenge, OpenBookQA, and Sports, their FF-HARD scores fall by 33.9, 40.4, and 22.4 points relative to the corresponding source files. Efficacy remains only 20.5–30.8%, while specificity stays at 99.2–100%, indicating that the intervention is highly conservative rather than broadly destructive. The opposite pattern appears on StrategyQA when our LR is substantially higher than the available source LR: FF-HARD rises by 61.7 points for LLaMA-3-8B and 40.0 points for Mistral-7B, but specificity falls to 80.2% and 65.6%, respectively. These are not clean gains in faithfulness; they reflect a stronger and less localized parameter intervention.

Appendix Figure 13 shows that FF-HARD and FF-SOFT move together under large changes in intervention strength. The joint behavior of efficacy, specificity, and answer-probability shift is therefore necessary for interpretation: an answer flip is informative only when the target step is successfully suppressed and unrelated behavior remains stable.

Case-level stability. Appendix Figure 14 adds

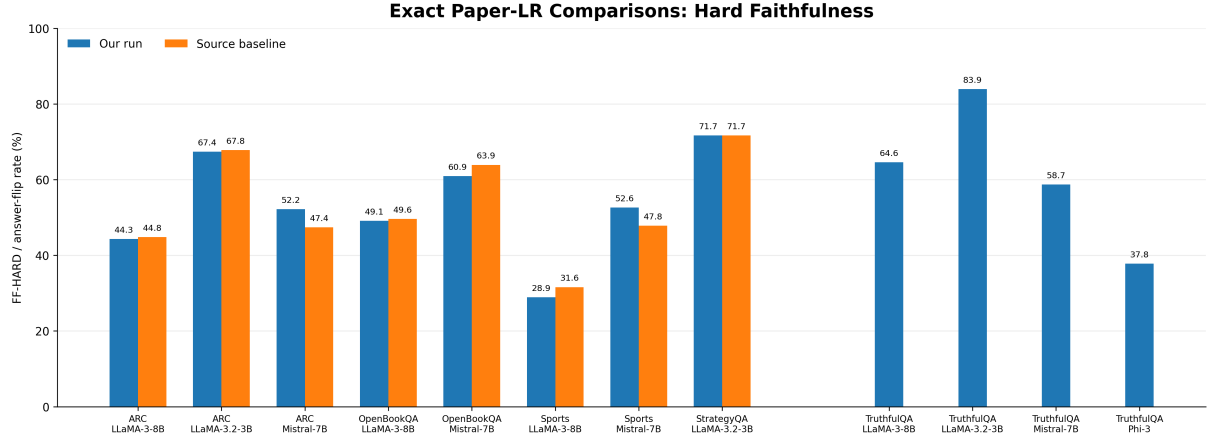


Figure 2: Exact-paper-LR reproduction at the question level. For the eight paired groups, blue bars show our FF-HARD and orange bars show the source baseline. **The original authors did not provide source-result data for TruthfulQA, so those four bars are excluded from the paired reproduction statistics.** Sports/LLaMA-3.2-3B is omitted because its local mainline run is incomplete.

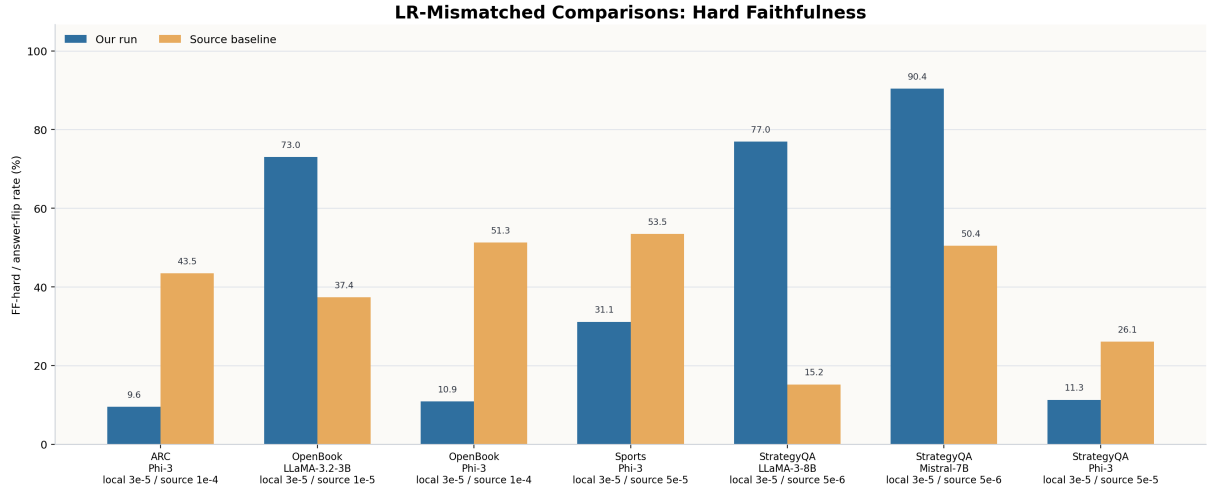


Figure 3: FF-HARD for the seven source-paired groups whose local and source result files use different learning rates. The direction of the gap follows intervention strength: weaker local updates tend to under-produce flips, whereas stronger updates can create many additional flips.

a stricter perspective than aggregate rates. Among source-paired runs whose local configuration uses the registry paper LR, 19% of aligned steps flip in both runs and 68% flip in neither, leaving 13% with a one-sided flip. When the local LR is higher than the registry setting, 36% of aligned steps flip only in our run. Thus, matched aggregate scores should be interpreted as reproducibility of the measurement distribution, not exact replication of every causal attribution.

3.4 Chinese Reproduction: Qwen3 + C-Eval

We extend FUR beyond English by applying it to Chinese multiple-choice reasoning on C-Eval (Huang et al., 2023). This experiment differs from the English reproduction along three axes

at once: the language is Chinese, the model family is Qwen3, and the benchmark consists of Chinese exam-style questions across many disciplines. Therefore, we treat the result as a cross-setting reproduction rather than a clean language-only ablation.

Setup. We evaluate Qwen3-8B and Qwen3-4B with visible Chinese CoTs. The Qwen chat template is used with native thinking disabled, so the rationale analyzed by FUR is the emitted Chinese text rather than hidden deliberation. CoTs are segmented by Chinese punctuation into sentence-like steps. All main runs use seed 1001, learning rate 10^{-5} , sentence-level stepwise NPO+KL, FF2/down-projection updates, and 20 held-out ex-

amples for specificity. The implementation record marks `pos=False`; unlike the English pipeline, it does not depend on English spaCy content-word POS tags.

Metric	Value
Faithful instances (FF-HARD)	21 / 230 = 9.13%
Per-step answer flips	25 / 498 = 5.02%
Mean efficacy	71.50%
Mean specificity	96.84%
Mean answer mass shift	0.0336

Table 2: Chinese FUR: Qwen3-8B on C-Eval. Unlearning is effective (71.5%) and specific (96.8%), but only 9.1% of instances show hard answer flips.

Table 2 summarizes the main results. The Qwen3-8B endpoint achieves high efficacy (71.50%) and specificity (96.84%), confirming that the NPO+KL intervention can suppress Chinese reasoning steps without broad same-task collapse. However, only 21 of 230 instances (9.13%) exhibit any hard answer flip, and only 25 of 498 step-level interventions (5.02%) flip the answer. Within this Qwen3 + C-Eval setting, most visible Chinese CoT steps are therefore not individually necessary for the direct answer. This does not imply that Chinese reasoning is inherently less faithful; model family, task distribution, step segmentation, and unlearning strength all differ from the English reproduction.

Strong Positive Cases. Despite the low overall flip rate, we observe clear cases of faithful steps in which unlearning a single reasoning step transfers nearly all probability mass from one option to another (e.g., a shift of 1.0000 from option C to B and a shift of 0.99994 from option A to B) while maintaining high specificity. These cases provide the strongest evidence of parametric faithfulness in the Chinese setting.

Model Size and Intervention Strength. Table 3 separates the matched five-epoch comparison from the extended-budget run. Under the directly comparable $n = 250, E = 5$ setting, Qwen3-4B has much higher efficacy than Qwen3-8B (74.81% vs. 63.12%) and slightly higher specificity (98.54% vs. 96.78%). The hard faithfulness increase is much smaller: 8.47% vs. 7.92%. On the common eligible subset of 158 questions, FF-HARD is 7.59% for Qwen3-4B and 6.96% for Qwen3-8B. Thus, the most stable conclusion is not that the smaller model is substantially more faithful, but that the same learning rate imposes a stronger local inter-

Run	Eff	Spec	FF-HARD	Step FF	Max FF-SOFT
Qwen3-8B ($n = 250, E = 5$)	63.12	96.78	7.92%	4.68%	6.59
Qwen3-4B ($n = 250, E = 5$)	74.81	98.54	8.47%	7.01%	7.19
Qwen3-4B ($n = 220$, through E5)	79.70	97.94	11.69%	9.32%	9.05

Table 3: Chinese reproduction under a unified five-epoch budget. The 4B model is easier to unlearn, but the increase in question-level FF-HARD is small under the matched $n = 250, E = 5$ setting.

Run	End E	Eff	Spec	FF-HARD(all)	FF-HARD(agree)	Mean Soft
Qwen3-8B ($n = 250$)	5	71.50	96.84	9.13%	6.93%	3.36
Qwen3-4B ($n = 250$)	5	81.50	98.56	9.57%	7.91%	4.38
Qwen3-4B ($n = 220$)	10	90.25	98.04	12.00%	10.39%	6.08

Table 4: Endpoint Chinese FUR results. The $n = 220$ 4B run uses ten epochs, so it should be interpreted as an intervention-strength extension rather than a pure size comparison.

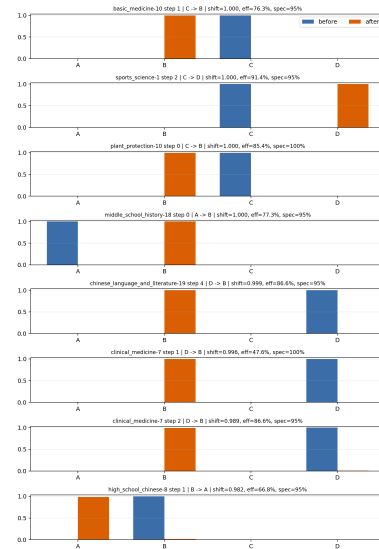


Figure 4: Top answer-flip examples from the Chinese reproduction. Each bar shows the redistribution of probability from the original answer to the new answer after one reasoning step is unlearned.

vention on it.

Table 4 reports each run at its own final checkpoint. The extended Qwen3-4B run reaches 90.25% efficacy and 12.00% FF-HARD after ten epochs, but the gain is modest after epoch 5: from epoch 5 to epoch 10, efficacy rises by 1.26 points and FF-HARD(all) rises by 1.00 point. The trajectory saturates after roughly epoch 6, which reinforces a central methodological lesson of FUR: stronger unlearning can reveal additional faithfulness signal, but beyond a point it mostly increases control strength rather than qualitatively changing the conclusion.

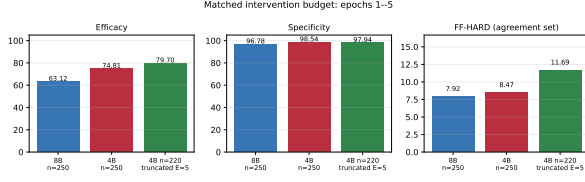


Figure 5: Chinese C-Eval diagnostics. Left: unified five-epoch comparison across Qwen3 runs. Right: endpoint trajectories, highlighting that the ten-epoch 4B extension saturates after the middle checkpoints.

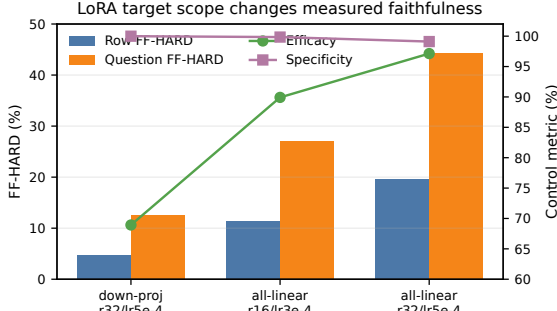


Figure 6: Within the tested LoRA configurations, target scope has the largest observed effect. Expanding adapters from down-projection-only to all linear layers improves efficacy and increases question-level FF-HARD from 12.61% to 44.35%, while specificity remains above 99%.

4 Extension Experiments

4.1 LoRA-Based Lightweight Intervention

Motivation. Full-parameter unlearning is computationally expensive and requires a separate model checkpoint for every unlearned step. We investigate whether Low-Rank Adaptation (LoRA) (Hu et al., 2022) provides a lightweight alternative that preserves the faithfulness measurement protocol while reducing computational cost.

Setup. We use Phi-3-mini-4k-Instruct on ARC-Challenge with stepwise NPO+KL and compare three LoRA configurations: down_proj only (rank 32, following the original FF2-only update scheme), all-linear with moderate rank/learning rate (rank 16, $\alpha = 32$, $\text{lr} = 3 \times 10^{-4}$), and all-linear with stronger settings (rank 32, $\text{lr} = 5 \times 10^{-4}$).

Results. Figure 6 presents the main comparison. Within the three tested configurations, **adapter target scope** has the largest observed effect on measured FF-HARD. The down_proj-only LoRA achieves only 12.61% question-level FF-HARD despite the same rank and learning rate. Expanding LoRA to all linear layers increases question-level FF-HARD to 44.35% while maintaining 99.11%

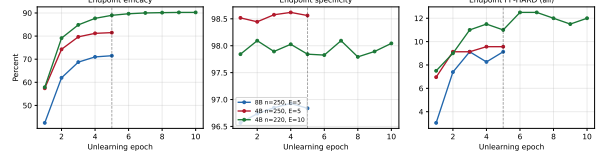


Figure 7: First- k forgetting is strongly front-loaded. Forgetting the first four target-step tokens recovers most of the question-level flip signal, while larger k mainly improves efficacy.

specificity. This pattern is consistent with answer-relevant information being accessible through a broader parameter subspace, but it should not be read as a direct localization result because changing the target scope also changes the intervention itself.

Ablation Studies. In addition to the main target-scope comparison, we conduct a systematic sweep of objective variants and mechanistic diagnostics. All experiments use Phi-3-mini on ARC-Challenge with all-linear LoRA of rank 32 and $\text{lr} = 5 \times 10^{-4}$.

4.1.1 First- k Forgetting Sweep

We restrict the forget objective to the first k tokens of the target CoT step to test whether disrupting only the step prefix blocks downstream reasoning. Figure 7 shows question-level results for $k \in \{1, 2, 4, 8, 16\}$ (940 rows across 230 questions per condition).

The behavioral effect is strongly front-loaded. At $k = 4$, the method already recovers most of the observed FF-HARD (40.00% vs. 44.78% at $k = 16$, full-step equivalent), whereas performance approaches saturation at $k \geq 8$. This result supports a “tip-of-the-pen forgetting” hypothesis: disrupting the initial tokens of a reasoning step is often sufficient to block the remaining reasoning trajec-

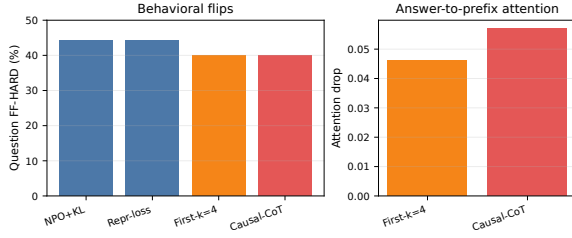


Figure 8: Mechanistic changes and answer flips are not interchangeable. Causal-CoT changes answer-to-prefix attention more than first- k forgetting but does not improve question-level FF-HARD.

tory, although later tokens still provide a marginal unlearning gain.

4.1.2 Representation-Loss and Causal-CoT Objectives

We evaluate two experimental objective variants against the NPO+KL baseline:

- **Repr-loss:** Adds a layer-weighted hidden-state similarity penalty between the target model and the frozen reference on the forget step. The objective moves internal representations away from the original reasoning trajectory rather than changing only token probabilities.
- **Causal-CoT:** A mechanism-aware objective that explicitly penalizes attention from answer-position tokens to the target-step tokens to block internal information routing through the unlearned step.

Figure 8 summarizes the comparison. Repr-loss is neutral because it neither improves nor degrades any metric relative to the NPO+KL baseline. Causal-CoT achieves its intended mechanistic effect, the strongest negative shift in attention from answer tokens to the unlearned step, but does not produce more answer flips. Question-level FF-HARD drops from 44.35% (baseline) to 40.00% (Causal-CoT), while specificity declines from 99.11% to 98.41%.

Interpretation. Greater mechanistic change does not necessarily improve answer-flip behavior. Causal-CoT successfully blocks internal attention routing through the unlearned step, but the model appears to compensate through alternative pathways and retains the final answer. The separation between mechanistic and behavioral signals warrants caution in interpretability research: attention-

level interventions can change internal computation without affecting downstream outputs.

4.1.3 Critical-Step Locus

We isolate the steps that cause answer flips under the strongest all-linear LoRA baseline and reevaluate the interventions on these critical steps. The critical-step runs (41 rows, 29 questions) confirm that reasoning steps differ in importance. The strongest intervention effects are concentrated in a small subset of steps, on which the baseline achieves 100% efficacy and 99.3% specificity. The same pattern holds for the repr-loss and Causal-CoT variants at the critical-step level, where repr-loss again does not improve the final flip rate over the baseline.

4.2 Enhanced FUR: Selective Top-1 Unlearning

Motivation. Full FUR unlearns every segmented CoT step and therefore requires 431 step-runs for 80 instances in our setting. We test whether unlearning only one selected step (top-1) can recover the faithful cases identified by full FUR at substantially lower computational cost. This extension is motivated by a practical bottleneck of PFF: the method is conceptually clean, but its cost scales linearly with the number of generated reasoning steps and requires a separate unlearning/evaluation trajectory for each step.

Setup. We compare three strategies on LLaMA-3.2-3B-Instruct with OpenBookQA (100 CoTs, 80 unlearning targets, 20 held-out): (1) **Full FUR:** unlearn all segmented CoT steps; (2) **Last-step top-1:** unlearn only the final reasoning step; (3) **Random top-1:** unlearn one randomly selected step.

Method	Rows	Faithful Inst.	Step Flip	Eff	Spec	Cost Red.
Full FUR	431	53/80 = 66.25%	28.31%	89.94	94.51	0.00%
Last top-1	78	22/78 = 28.21%	28.21%	88.96	94.74	81.90%
Random top-1	71	21/71 = 29.58%	29.58%	89.65	93.73	83.53%

Table 5: Selective FUR comparison on OpenBookQA. Full FUR finds 66.25% faithful instances; top-1 variants preserve high efficacy and specificity while reducing unlearning rows by more than 80%.

Results. Table 5 shows that Full FUR identifies 66.25% of instances as parametrically faithful. The Wilson 95% interval for this rate is [55.4, 75.7], which confirms that OpenBookQA + LLaMA-3.2-3B has a much denser faithfulness signal than the Chinese C-Eval setting. Both top-1 strategies substantially reduce cost (81.90% for last-step and

Selector	Found / Full Faithful	Recovery@1	Coverage	Step-Hit@1	Precision
Last-step top-1	22 / 53	41.51%	97.50%	17.21%	26.92%
Random top-1	21 / 53	39.62%	88.75%	18.85%	32.39%

Table 6: Recovery against Full FUR. Top-1 methods recover about 40% of Full FUR faithful instances while running less than one fifth of the unlearning jobs.

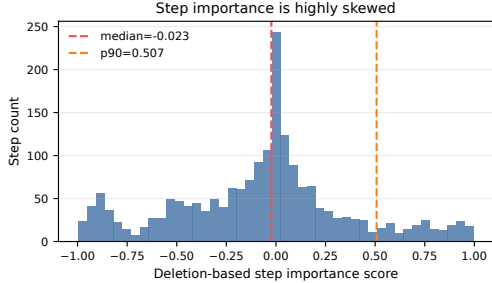


Figure 9: Deletion-based step importance on ARC-Challenge is highly skewed. Most steps have small or negative scores, whereas a small tail produces large drops in answer probability. This distribution explains why simple top-1 selection misses many faithful steps.

83.53% for random), while keeping efficacy around 89% and specificity around 94%. The selective rows have step flip rates close to Full FUR, but this is partly a denominator effect: each selected instance contributes only one tested step.

Recovery–Cost Tradeoff. Table 6 gives the key efficiency result. Last-step top-1 recovers 22 of the 53 Full-FUR faithful instances (41.51%), and random top-1 recovers 21 (39.62%). The last-step heuristic is therefore not clearly better than the random baseline. Random top-1 even has slightly higher selected-step precision (32.39% vs. 26.92%), suggesting that final CoT sentences are not a reliable proxy for causal importance. This result supports selective FUR as a cost-reduction direction, but the missing component is a stronger step ranker, such as deletion sensitivity, verifier scoring, answer-mention detection, or a hybrid uncertainty criterion.

Strong Answer-Transfer Cases. Full FUR also surfaces cases in which a single unlearned step moves almost all probability mass from the original answer to another option. For example, instance 508 changes from D to A with an original-answer drop of 0.9272 and 91.56% efficacy; instance 266 changes from D to C with a drop of 0.9256 and 96.09% efficacy. Such cases are the most persuasive qualitative evidence that FUR can identify reasoning steps that are not merely verbal decorations. At the same time, the efficacy–answer-shift scatter

shows that high target-step suppression alone is insufficient: many steps are successfully forgotten without crossing the answer decision boundary.

4.3 Measuring Tree-of-Thought Faithfulness

Motivation. We generalize PFF from linear CoT to Tree-of-Thought (ToT) (Yao et al., 2023) reasoning and test whether multi-path deliberation reveals stronger dependence on model parameters than a single reasoning path. CoT produces one rationale, whereas ToT samples or searches over multiple candidate paths before choosing an answer. If a selected ToT path is a better match to the model’s internal evidence, unlearning its steps should cause larger answer-probability changes than unlearning a greedy CoT. Conversely, if ToT mainly improves surface-level search without changing the parametric basis of the answer, FF-HARD should remain unchanged.

Setup. We compare two reasoning arms on LLaMA-3.2-3B-Instruct: (1) **CoT**: a single visible reasoning path (standard FUR baseline); (2) **ToT-selected**: five sampled reasoning paths, from which we select the path with the highest confidence for unlearning. Both arms use the same NPO+KL protocol, FF2-only updates, and 5 unlearning epochs. We evaluate on OpenBookQA (50 forget, 20 retain) and ARC-Challenge (100 forget, 20 retain). Before the full run, we compare `sample-select` and `beam-prune` on a validation subset. The best-of-five `sample-select` procedure is selected for both datasets because it gives higher validation accuracy while maintaining five complete candidate paths.

Dataset	Arm	Eff	Spec	FF-HARD	FF-SOFT
OpenBookQA	CoT	81.2	95.4	72.5	50.2
OpenBookQA	ToT	81.8	95.1	71.8	55.3
ARC-Challenge	CoT	75.8	95.6	68.8	51.3
ARC-Challenge	ToT	79.6	95.4	72.2	58.5

Table 7: CoT vs. ToT faithfulness. ARC-Challenge shows the clearest advantage of ToT: FF-HARD +3.4 and FF-SOFT +7.3, with almost unchanged specificity.

Results. Table 7 shows that the effect of multi-path reasoning depends on the dataset. On ARC-Challenge, ToT-selected improves FF-HARD by 3.4 points and FF-SOFT by 7.3 points with minimal specificity change. On OpenBookQA, ToT improves FF-SOFT (+5.2) but does not increase strict FF-HARD (-0.7). Thus, ToT can increase the *degree* to which the original answer probability is

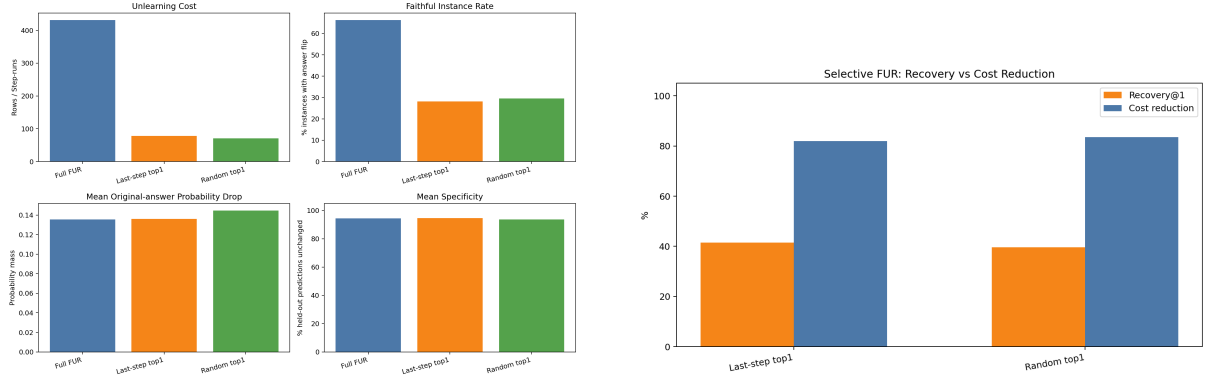


Figure 10: Enhanced FUR diagnostics. Left: Full FUR has much higher instance-level recall because it tests all steps. Right: both selective top-1 variants remove more than 80% of unlearning jobs but recover only about 40% of Full-FUR faithful cases.

Dataset	Common Agree N	CoT FF-HARD	ToT FF-HARD	Δ
OpenBookQA	37	70.3	70.3	0.0
ARC-Challenge	62	64.5	67.7	+3.2

Table 8: Paired CoT-ToT comparison on common agreement-eligible questions. The ARC advantage remains after controlling for the agreement subset.

supported by selected reasoning steps even when it does not change the binary flip rate.

Because CoT and ToT can change the set of questions that pass the direct/CoT agreement filter, Table 8 provides a stricter paired comparison. OpenBookQA is tied on the common subset, while ARC-Challenge retains a +3.2 point FF-HARD advantage for ToT. This paired result is smaller than the unpaired table but more interpretable: it suggests that multi-path selected reasoning is most helpful when the task requires multi-hop scientific discrimination, not merely when it changes which questions are eligible for evaluation.

This difference suggests that **multi-path reasoning exposes stronger parameter dependence on datasets requiring scientific reasoning** (ARC-Challenge) than on commonsense datasets (OpenBookQA), where reasoning may remain more “surface-level” regardless of the number of paths.

4.4 Distilled CoT via DeepSeek

Motivation. We test whether one supervised fine-tuning (SFT) pass on teacher-revised CoTs can better align generated reasoning with the parametric knowledge of the model. The hypothesis is intuitive: if a stronger teacher rewrites student rationales into shorter, correct, and less noisy explanations, then SFT on those rationales may make the student’s visible CoT more tightly coupled to

the answer-supporting knowledge in its parameters. FUR gives a way to test this hypothesis beyond accuracy.

Setup. We use LLaMA-3.2-3B-Instruct as the student, with DeepSeek-V4-Pro as the teacher. The teacher revises 2,000 student-generated OpenBookQA CoTs to improve conciseness and answer consistency (92.6% acceptance rate, 1,852 SFT examples). SFT uses LoRA (down_proj only, rank 32, $\alpha = 64$, 1 epoch, $\text{lr} = 10^{-4}$). We then apply the same FUR protocol to both Base and SFT models on a shared 100-question test split. All teacher revisions are generated from the training split, while FUR evaluation uses an independent OpenBookQA test subset with 100 target questions and 20 held-out retain questions.

Model	Dir Acc	CoT Acc	Agree	Steps	Eff	Spec
Base	70.00	73.00	82.00	6.37	92.58	94.04
SFT	73.00	77.00	85.00	3.12	97.62	98.79

Table 9: Distilled CoT: SFT improves accuracy and FUR quality. FF-SOFT +1.97 pp, FF-HARD −4.85 pp; both n.s. on 75 paired questions.

Results. Table 9 shows that SFT on DeepSeek-revised CoTs improves several desirable properties: CoTs become more concise ($6.37 \rightarrow 3.12$ steps on average), the quality of FUR unlearning improves (higher efficacy and specificity), and both direct and CoT-conditioned accuracy increase. The number of valid FUR steps falls from 495 to 311, reflecting the strong compression effect of teacher revisions.

Table 10 shows the core faithfulness tension. Under each model’s own agreement-eligible questions, FF-SOFT increases from 54.09 to 56.07, but FF-

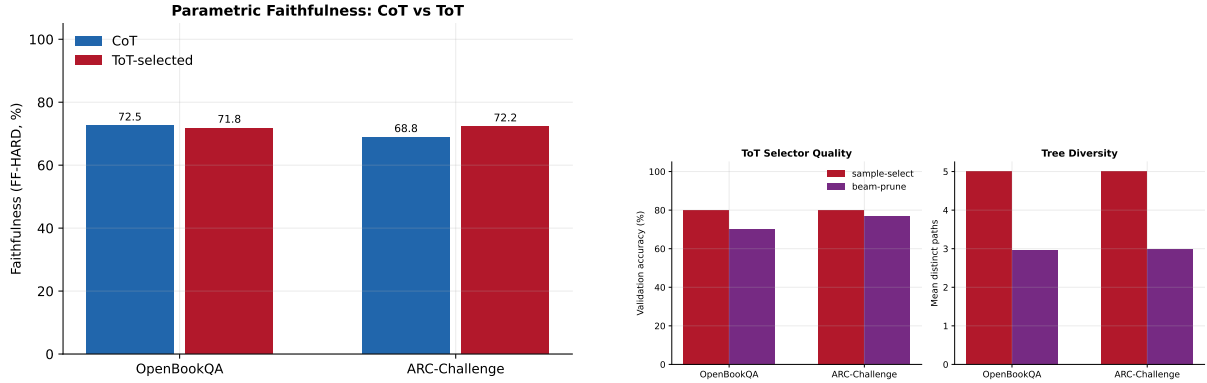


Figure 11: Tree-of-Thought diagnostics. Left: FF-HARD comparison between CoT and selected ToT. Right: validation-based mode selection; sample-select is used for the final ToT runs.

Model	Agree N	FF-HARD Agree	FF-SOFT Agree	FF-HARD All	FF-SOFT All
Base	82	70.73	54.09	72.00	51.97
SFT	85	65.88	56.07	67.00	52.87

Table 10: Faithfulness after DeepSeek-revision SFT. FF-SOFT increases slightly, whereas FF-HARD decreases under the model-specific agreement filter.

HARD decreases from 70.73 to 65.88. On the stricter set of 75 common agreement questions, FF-HARD decreases by 4.00 pp (95% bootstrap CI $[-16.00, 8.00]$) and FF-SOFT increases by 1.12 pp (CI $[-7.95, 9.95]$); neither change is statistically significant.

Interpretation. One-shot distilled-CoT SFT improves compactness and FUR control quality but does not establish a statistically stable gain in faithfulness. This result may indicate that (a) more SFT data or multiple epochs are required, (b) the revisions of the teacher, although more concise, do not necessarily strengthen the causal connection between the CoT and internal reasoning, (c) shorter CoTs reduce the number of opportunities for an “exists a flipping step” FF-HARD event, or (d) the inherent variability of the FUR protocol obscures small improvements in faithfulness. Another confound is that both SFT and FUR act on the `down_proj` parameter subspace; improved efficacy may partly reflect increased plasticity of that subspace rather than a deeper causal alignment between rationale and answer.

5 Discussion

5.1 The Nature of Parametric Faithfulness

Our results do not support a single, uniformly sparse FF-HARD regime. Under exact-paper-LR paired settings, question-level FF-HARD ranges

from 28.9% to 71.7%. By contrast, Qwen3-8B on C-Eval yields 9.13%, and conservative Phi-3 interventions remain near 10% on ARC-Challenge and OpenBookQA. Measured faithfulness is therefore conditional on the model, dataset, rationale segmentation, and intervention strength.

A hard flip remains strong positive evidence that the unlearned step carried answer-relevant parametric information. The absence of a flip is less decisive: the step may be redundant, the answer may be supported by an alternative path, or the intervention may be too weak. FUR is best interpreted as a controlled lower-bound probe whose validity depends on efficacy and specificity, rather than as a complete census of faithful reasoning steps.

5.2 Hyperparameter Sensitivity as a Methodological Concern

The reproduction audit quantifies the effect of optimization control. Exact-learning-rate pairs differ by only 2.07 FF-HARD points on average, whereas LR-mismatched pairs differ by 35.55 points. The largest positive gaps coincide with substantial specificity loss, while the largest negative gaps coincide with low efficacy. Case-level overlap is also incomplete even when aggregate rates are close.

These observations motivate three reporting requirements for unlearning-based faithfulness studies:

- report the complete dataset–model-specific optimization configuration and distinguish matched from mismatched comparisons;
- calibrate intervention strength jointly against efficacy and a pre-specified specificity floor, rather than maximizing FF-HARD alone;

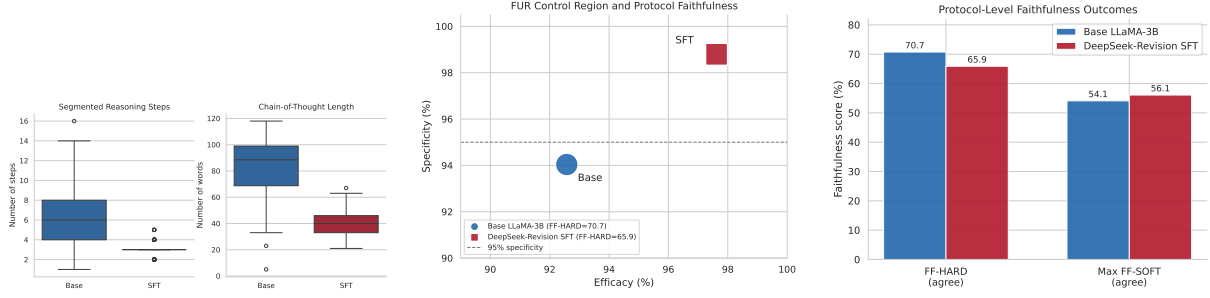


Figure 12: DeepSeek-SFT diagnostics. SFT compresses CoTs and improves FUR control quality, but the resulting faithfulness metrics split: soft probability mass decreases slightly more, whereas hard flips do not increase.

- report both aggregate faithfulness rates and case-level overlap, because similar totals can arise from different sets of flipping steps.

5.3 LoRA Target Scope and the Locus of Faithful Reasoning

The substantial effect of expanding LoRA targets from down-projections to all linear layers (12.6% → 44.3% question-level FF-HARD) shows that measured FF-HARD depends strongly on intervention scope. The result is consistent with answer-relevant information being distributed across attention and FFN components, but it does not by itself localize that information or establish that the larger score is a more accurate estimate of faithfulness. Parameter scope should therefore be reported and calibrated as part of the measurement protocol rather than treated as a neutral implementation detail.

5.4 Scaling Faithfulness Measurement

The computational cost of full-stepwise FUR (431 step-runs for 80 instances) limits wider adoption. Our selective FUR experiments show that computational cost can be reduced by over 80% while preserving approximately 40% recovery of faithful cases. Together with lightweight LoRA-based intervention, scalable faithfulness measurement requires: (1) efficient parameter interventions (LoRA instead of full fine-tuning), (2) intelligent step selection (learned verifiers instead of naive heuristics), (3) early-stopping protocols that terminate unlearning once sufficient efficacy is reached.

5.5 Multi-Path Reasoning and Faithfulness

Our ToT extension provides initial evidence that multi-path deliberation can reveal stronger parameter dependence, particularly in scientific reasoning tasks. This result suggests a connection between reasoning diversity and measurable faithfulness:

when a model considers multiple reasoning paths, the selected path may be more closely coupled with internal parametric knowledge. However, this effect depends on the dataset. It is stronger on ARC-Challenge and absent on OpenBookQA, indicating that task characteristics mediate the relationship between reasoning structure and faithfulness.

6 Conclusion

We presented a comprehensive reproduction and multi-axis extension of the Parametric Faithfulness Framework and its FUR instantiation. FUR provides a useful causal test of whether a stated CoT step is supported by answer-relevant parametric information, but the resulting score is meaningful only under a controlled intervention.

The English audit supports aggregate reproducibility when configurations are directly comparable: across eight exact-paper-LR pairs, the mean absolute FF-HARD gap is 2.1 percentage points and the maximum gap is 4.8 points. In contrast, seven LR-mismatched pairs differ by 35.5 points on average, with weak interventions suppressing flips and aggressive interventions reducing specificity. Close aggregate totals do not amount to exact case-level replication, because the sets of flipping questions and steps can still differ. Beyond the reproduction, Qwen3 on C-Eval shows a sparse hard-flip signal (9.13%); all-linear LoRA substantially increases measured FF-HARD relative to down-projection-only adapters; selective FUR removes more than 80% of unlearning jobs while recovering about 40% of full-FUR faithful cases; ToT strengthens the signal on ARC-Challenge; and one-pass distilled-CoT SFT improves compactness and control quality without a statistically stable FF-HARD gain.

Taken together, these results position FUR as a diagnostic protocol rather than a model-intrinsic

scalar. Reliable comparisons require matched optimization and joint reporting of efficacy, specificity, hard flips, soft probability movement, and case-level stability.

7 Limitations

Our study inherits the limitations of the PFF/FUR framework and introduces additional limitations:

Scope restriction: FUR analyzes only samples for which CoT and no-CoT answers agree. This restriction reduces contextual interference but excludes cases in which CoT changes the answer.

Unlearning imperfection: Machine unlearning may erase information incompletely or imprecisely, which may lead to underestimation of faithfulness.

Closed-source incompatibility: FUR requires parameter access, making it inapplicable to API-only models (GPT-4, Claude, etc.).

Task limitation: All experiments use MCQA tasks; the method has not been validated on long-form generation, mathematical reasoning, or code generation.

Computational cost: Full-stepwise FUR scales poorly to long reasoning chains, although our selective and LoRA-based extensions partially reduce this cost.

Reproduction coverage and stability: The exact-configuration paired audit covers eight of the 20 intended dataset-model groups and uses a single random seed. The 2.07-point mean gap therefore characterizes the available matched comparisons rather than the full grid or across-run variability. In addition, similar aggregate FF-HARD rates can conceal different instance-level flips.

Chinese generalizability: Our Chinese experiments use one model family (Qwen3) and one benchmark (C-Eval); broader evaluation in Chinese is required.

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A Experimental Details

This appendix consolidates implementation settings recorded in the repository code, run scripts, and active experiment registry. Unless noted otherwise, FUR uses sentence-level stepwise unlearning with NPO+KL and content-word filtering.

A.1 Core FUR Configuration

Table 11: Shared implementation settings for the principal FUR reproduction runs.

Setting	Value
Unlearning objective	NPO+KL
NPO β / KL coefficient	0.1 / 1.0
Optimizer / schedule	AdamW / linear decay, no warmup
Random seed / CoT temperature	1001 / 0
Segmentation / protocol	Sentence-level / stepwise
CoT cap / specificity split	250 / 20 examples
Retain CoTs / batch size	4 / 1
Epochs / evaluation interval	5 / every epoch
Forget-token filter	Nouns, verbs, adjectives, numerals
Updated parameters	FFN/MLP down_proj

The implementation uses a frozen copy of the original model as π_{ref} . CUDA runs load model weights in bfloat16. The AdamW optimizer is applied only to trainable parameters, with a linear learning-rate schedule over all unlearning steps and zero warmup steps.

Table 12: Dataset-specific learning rates stored in the reproduction registry.

Dataset	Phi-3	LLaMA-3-8B	LLaMA-3.2-3B	Mistral-7B
ARC-Challenge	1×10^{-4}	1×10^{-5}	3×10^{-5}	5×10^{-6}
OpenBookQA	1×10^{-4}	1×10^{-5}	3×10^{-5}	5×10^{-6}
BBH Sports	5×10^{-5}	5×10^{-6}	3×10^{-5}	3×10^{-6}
StrategyQA	5×10^{-5}	1×10^{-5}	3×10^{-5}	5×10^{-6}
TruthfulQA	5×10^{-5}	1×10^{-5}	3×10^{-5}	3×10^{-6}

A.2 LoRA and Objective Extensions

All active LoRA scope experiments use Phi-3-mini-4k-Instruct on ARC-Challenge, NPO+KL, seed 1001, five epochs, POS-based content-word filtering, four retain CoTs, and 20 held-out examples. LoRA dropout is 0. Each configuration covers 940 completed step-level rows from 230 source questions across two execution shards.

Table 13: Active LoRA target-scope configurations. An empty target-module list wraps all `nn.Linear` layers.

Configuration	Rank	α	LR	Targets
Down-projection	32	32	5×10^{-4}	down_proj
All-linear, moderate	16	32	3×10^{-4}	All linear layers
All-linear, strong	32	32	5×10^{-4}	All linear layers

A.3 Settings for Additional Study Branches

Chinese FUR: Qwen3-8B on C-Eval; 230 unlearning instances and 20 held-out examples; visible Chinese CoTs without native thinking mode; sentence-level stepwise NPO+KL.

Table 14: Parameters for the loss and diagnostic extensions.

Extension	Configuration
First- k forgetting	$k \in \{1, 2, 4, 8, 16\}$; all-linear LoRA, rank 16, $\alpha = 32$, $\text{lr} = 5 \times 10^{-4}$, five epochs
Representation loss	Weight 0.1; final 4 hidden layers; layer discount $\gamma = 0.9$; first 4 target tokens; automatic loss scaling
Causal-CoT	Auxiliary weight 0.1; margin 1.0; intervention and margin weights 1.0; <code>remove_step</code> counterfactual; correct-answer target; automatic loss scaling
Critical-step runs	Deletion-importance threshold 0.9; all-linear LoRA, rank 32, $\alpha = 32$, $\text{lr} = 5 \times 10^{-4}$
Mechanistic diagnostics	Up to 24 projection/QKV-style modules summarized during evaluation

The matched size comparison uses Qwen3-4B and Qwen3-8B with $n = 250$ and 5 epochs; the extended 4B run uses $n = 220$ and 10 epochs.

Selective FUR: LLaMA-3.2-3B-Instruct on OpenBookQA; 100 CoTs split into 80 unlearning targets and 20 held-out examples; full-step FUR compared with last-step and random single-step selection.

Tree-of-Thought: LLaMA-3.2-3B-Instruct; five sampled paths with the highest-confidence path selected for unlearning; NPO+KL, FF2-only updates, and 5 epochs; OpenBookQA uses 50 forget and 20 retain examples, whereas ARC-Challenge uses 100 forget and 20 retain examples.

Distilled CoT: LLaMA-3.2-3B-Instruct student and DeepSeek-V4-Pro teacher; 2,000 revised CoTs with 1,852 accepted SFT examples; down-projection LoRA with rank 32, $\alpha = 64$, $\text{lr} = 10^{-4}$, and 1 epoch; evaluation on a shared 100-question split.

A.4 Metric Aggregation

The repository distinguishes step-level rows from source questions. Row-level FF-HARD is the percentage of individual unlearned steps whose final answer differs from the initial answer. Question-level FF-HARD counts a question as flipped when at least one associated step flips the final answer. FF-SOFT is the mean reduction in probability assigned to the initial answer. Efficacy is the relative reduction in the probability of the target CoT step, and specificity is prediction agreement before and after unlearning on held-out same-task examples. Unless a result is explicitly labeled as question-level, the summary scripts aggregate over completed step-level rows.

A.5 English Reproduction Audit

Tables 15–17 give the numerical values used in Section 3. FF-HARD is question-level; FF-SOFT is the mean percentage-point reduction in the initial-answer probability. “O/S” denotes our run/source baseline; the final table lists runs without a paired source result.

The remaining intended group, Sports/LLaMA-3.2-3B, has only one question in each local mainline output file. A larger file exists under the add-mistake/minimal-mistake branch, but it does not contain comparable mainline unlearning results and is therefore excluded.

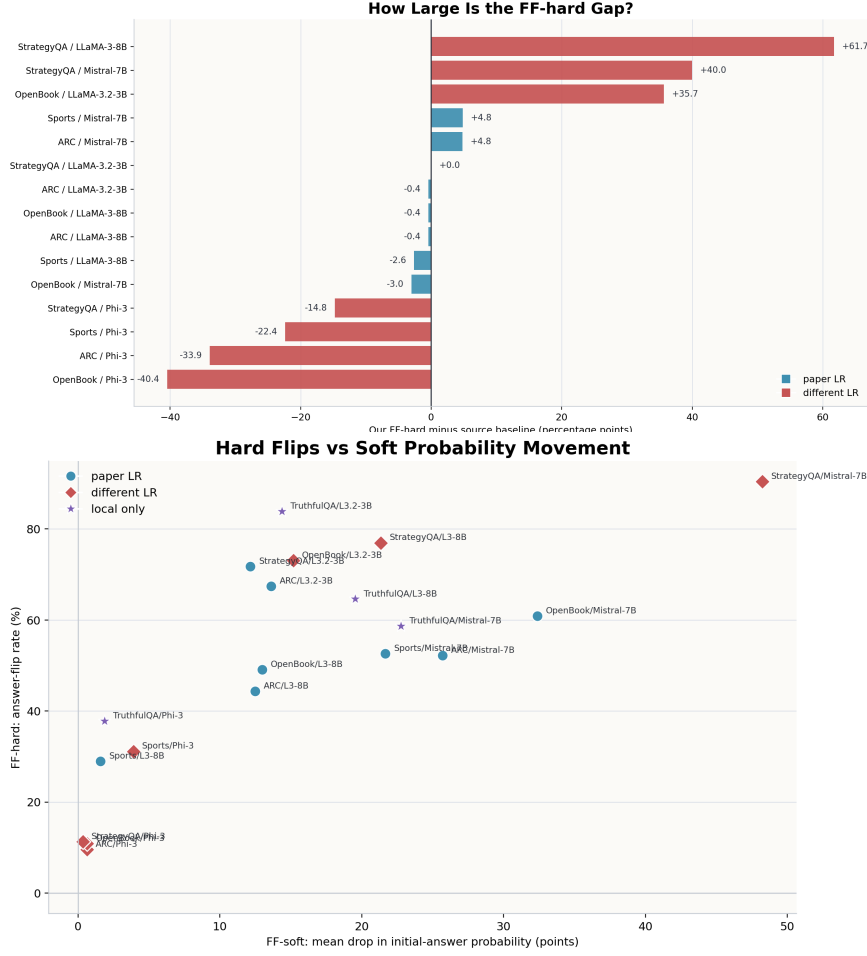


Figure 13: Reproduction diagnostics. Top: signed FF-HARD gap (our run minus source baseline); exact-paper-LR pairs remain within approximately five points, whereas LR-mismatched pairs produce much larger deviations. Bottom: relationship between question-level FF-HARD and FF-SOFT; aggressive mismatches occupy the high-shift, high-flip region, conservative mismatches cluster near zero FF-SOFT, and the unpaired TruthfulQA points are included for reference.

Table 15: Exact-paper-LR source-paired reproduction results.

Dataset	Model	FF-HARD O	FF-HARD S	Δ	FF-SOFT O/S
ARC	LLaMA-3-8B	44.3	44.8	-0.4	12.5 / 13.4
ARC	LLaMA-3.2-3B	67.4	67.8	-0.4	13.6 / 13.3
ARC	Mistral-7B	52.2	47.4	+4.8	25.7 / 23.8
OpenBookQA	LLaMA-3-8B	49.1	49.6	-0.4	13.0 / 13.3
OpenBookQA	Mistral-7B	60.9	63.9	-3.0	32.4 / 33.5
Sports	LLaMA-3-8B	28.9	31.6	-2.6	1.6 / 1.6
Sports	Mistral-7B	52.6	47.8	+4.8	21.7 / 19.7
StrategyQA	LLaMA-3.2-3B	71.7	71.7	+0.0	12.2 / 10.9

Table 17: TruthfulQA results from our runs at the registry learning rates. These values are not included in the paired reproduction statistics.

Model	Target IDs	FF-HARD	FF-SOFT	Specificity
LLaMA-3-8B	229	64.6	19.5	93.6
LLaMA-3.2-3B	230	83.9	14.4	85.8
Mistral-7B	230	58.7	22.8	99.2
Phi-3	230	37.8	1.9	93.3

Table 16: Source-paired comparisons with different learning rates. LR, FF-HARD, and specificity are shown as our run/source baseline.

Dataset	Model	LR O/S	FF-HARD O/S	Δ	Spec O/S
ARC	Phi-3	3e-5/1e-4	9.6 / 43.5	-33.9	100.0 / 99.4
OpenBookQA	LLaMA-3.2-3B	3e-5/1e-5	73.0 / 37.4	+35.7	95.2 / 99.4
OpenBookQA	Phi-3	3e-5/1e-4	10.9 / 51.3	-40.4	100.0 / 99.3
Sports	Phi-3	3e-5/5e-5	31.1 / 53.5	-22.4	99.2 / 96.4
StrategyQA	LLaMA-3-8B	3e-5/5e-6	77.0 / 15.2	+61.7	80.2 / 98.2
StrategyQA	Mistral-7B	3e-5/5e-6	90.4 / 50.4	+40.0	65.6 / 95.5
StrategyQA	Phi-3	3e-5/5e-5	11.3 / 26.1	-14.8	99.8 / 97.8

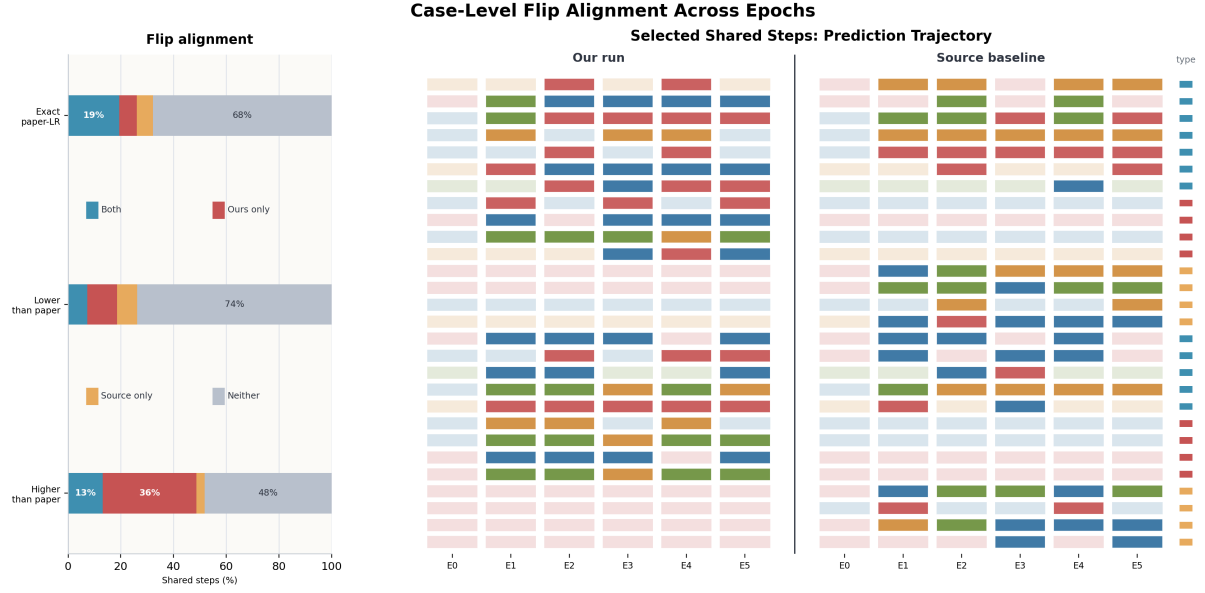


Figure 14: Case-level flip alignment across learning-rate classes. Our runs are grouped by their learning rate relative to the registry paper setting. The left panel separates aligned steps that flip in both runs, in only one run, or in neither; the right panels show selected answer trajectories across unlearning epochs. Aggregate FF-HARD agreement does not imply identical case-level interventions, and higher-than-paper local LRs create a large share of one-sided flips in our runs.

B Chinese Reproduction Details

Model	CoTs	Targets	Steps	Epochs	Agreement-eligible
Qwen3-8B	250	230	498	5	202
Qwen3-4B	250	230	344	5	177
Qwen3-4B	220	200	299	10	154

Table 18: Completeness audit for the Chinese C-Eval runs. The two $n = 250$ runs share target question IDs, but each model generates its own CoT and step segmentation.

Model	Common Eligible N	FF-HARD(1-5)	FF-HARD(E5)	Max FF-SOFT
Qwen3-8B	158	6.96%	6.33%	6.58
Qwen3-4B	158	7.59%	6.96%	6.58

Table 19: Paired Chinese comparison on questions where both Qwen3-8B and Qwen3-4B satisfy the initial direct-CoT agreement filter.

Teacher Filtering Status	Count	Rate
accepted	1852	92.60%
wrong teacher answer	88	4.40%
invalid JSON	35	1.75%
answer leak	13	0.65%
sentence count	10	0.50%
rationale length	2	0.10%

Table 20: DeepSeek teacher-revision filtering audit.

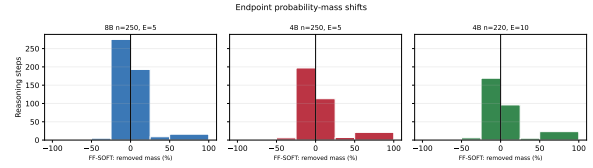


Figure 15: Final-checkpoint FF-SOFT distributions for the Chinese runs. The ten-epoch 4B run shifts more steps rightward, but this combines sample and intervention-budget changes.

C Selective FUR Details

Method	Faithful Rate	95% CI	Step Flip Rate	95% CI
Full FUR	66.25%	[55.4, 75.7]	28.31%	[24.3, 32.7]
Last-step top-1	28.21%	[19.4, 39.0]	28.21%	[19.4, 39.0]
Random top-1	29.58%	[20.2, 41.0]	29.58%	[20.2, 41.0]

Table 21: Wilson 95% confidence intervals for Enhanced FUR rates.

Dataset	Mode	Selected	Accuracy	Confidence	Paths	Vote Entropy
OpenBookQA	sample-select	Yes	80.0	100.0	5.00	0.320
OpenBookQA	beam-prune	No	70.0	98.9	2.97	0.106
ARC-Challenge	sample-select	Yes	80.0	100.0	5.00	0.328
ARC-Challenge	beam-prune	No	76.7	100.0	3.00	0.127

Table 22: Validation-based ToT procedure selection.

D Tree-of-Thought Details

E DeepSeek-SFT Details

ID	Step	Change	Correct	Drop	TV	Eff	Question
508	1	D $\rightarrow$ A	D	0.9272	0.9272	91.56	To improve health, what is a good strategy?
266	3	D $\rightarrow$ C	D	0.9256	0.9583	96.09	What is used for sensing visual things?
1189	1	B $\rightarrow$ A	A	0.9247	0.9247	92.04	What can feathers on Spheniscidae be used for?
1577	7	B $\rightarrow$ D	B	0.8997	0.9010	95.81	What animal eats plants?
1577	11	B $\rightarrow$ A	B	0.8939	0.8939	97.16	What animal eats plants?

Table 23: Strong probability-transfer examples from Full FUR on OpenBookQA. TV denotes total variation distance between answer distributions before and after the intervention.

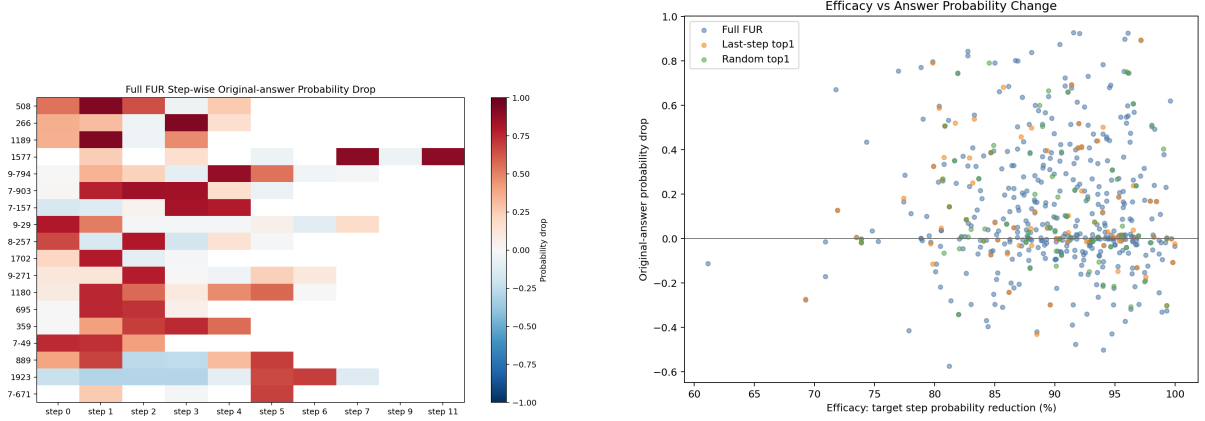


Figure 16: Selective FUR supporting diagnostics. Left: faithful steps are concentrated in a small number of high-salience instances. Right: high efficacy is necessary for interpretation but does not guarantee a large answer-probability shift.

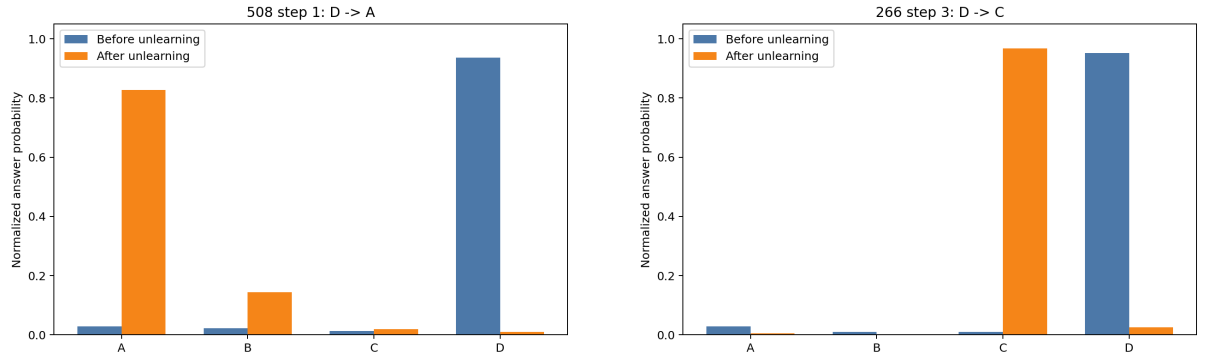


Figure 17: Two full-distribution answer-transfer examples from Enhanced FUR.

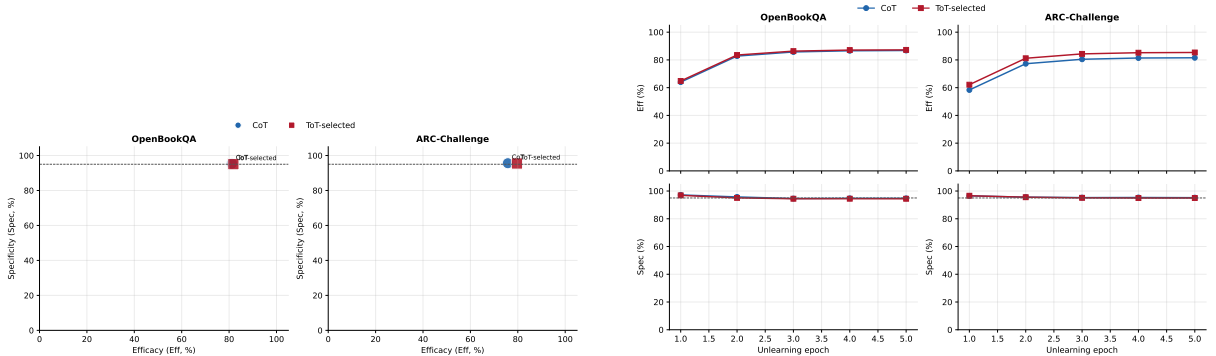


Figure 18: ToT control diagnostics. The ToT ARC run improves efficacy while maintaining nearly identical specificity, supporting the interpretation that the ARC faithfulness gain is not caused by broad model damage.

Metric	N	Base	SFT	Delta [95% CI]	McNemar (+/-; p)
Direct accuracy	100	70.00	73.00	+3.00 [-3.00, 9.00]	6/3; $p = 0.508$
CoT accuracy	100	73.00	77.00	+4.00 [-3.00, 11.00]	8/4; $p = 0.388$
Direct-CoT agreement	100	82.00	85.00	+3.00 [-5.00, 11.00]	10/7; $p = 0.629$
CoT steps	100	6.37	3.12	-3.25 [-3.83, -2.72]	–
CoT words	100	81.33	40.49	-40.84 [-45.50, -35.93]	–
FF-HARD (common agree)	75	68.00	64.00	-4.00 [-16.00, 8.00]	10/13; $p = 0.678$
FF-SOFT (common agree)	75	54.49	55.61	+1.12 [-7.95, 9.95]	–
Initial margin (common agree)	75	73.18	82.49	+9.30 [5.34, 13.38]	–

Table 24: Strict paired DeepSeek-SFT comparison. Compactness changes are stable, but faithfulness deltas are not statistically decisive.

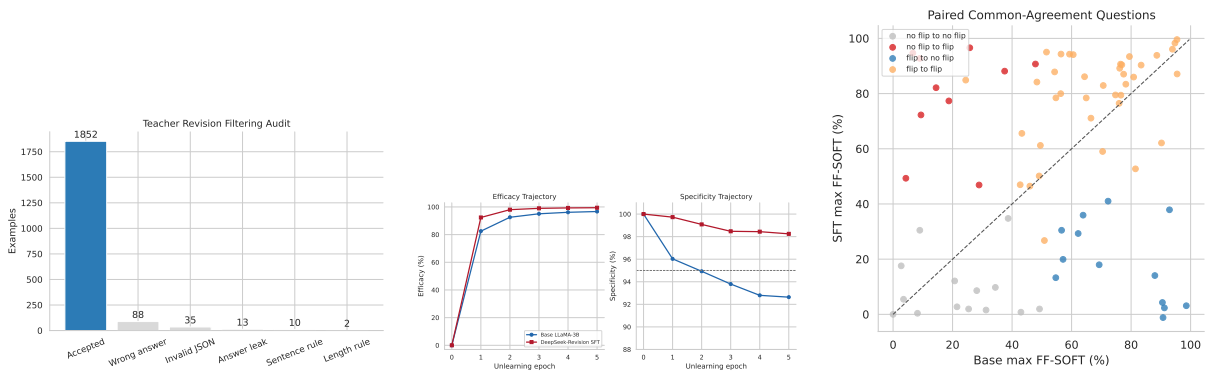


Figure 19: DeepSeek-SFT supporting diagnostics: teacher filtering, control trajectories, and paired FF-SOFT distribution.

F Artifact Availability

The report source and lightweight assets are stored in the project repository. Large result files, model outputs, and experiment artifacts are stored in the corresponding ModelScope dataset. The main paths used for this report are:

- **GitHub:** [project repository](#)
- **ModelScope:** [ModelScope dataset](#)
- **Chinese FUR:** `parametric-faithfulness-main-cn/report/`
- **Selective FUR:** `parametric-faithfulness-enhanced/result_enhanced/`
- **Tree-of-Thought:** `parametric-faithfulness-ToT/report/`
- **DeepSeek-SFT:** `parametric-faithfulness-SFT/report/`

The original authors' released results are available at the following links:

- Full final results: https://drive.google.com/file/d/1aCbj-Uk1-3kMYR6_yiLM6yxgpT40b5Y/view?usp=drive_link
- Ablation results: https://drive.google.com/file/d/1Qm1PlHa5PtB9ubtT9-IPZoCEepn9WQRJ/view?usp=drive_link